SAINT MICHAEL COLLEGE OF CARAGA

ONLINE REGISTRATION SYSTEM FOR THE CIVIL REGISTRAR'S OFFICE OF BUENAVISTA AGUSAN DEL NORTE

A Thesis Presented To
The Faculty of
College of Computing and Information Sciences

In Partial Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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CERTIFICATE OF TECHNOLOGY BASED ASSESSMENT 68
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ABSTRACT

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DEDICATION

This study is dedicated to our beloved families, whose unconditional love, patience, and unwavering support have been our constant source of strength and inspiration throughout this journey.

To our friends, advisers, and especially to our parents, who believed in our dreams and encouraged us to pursue our goals despite the challenges, your sacrifices and guidance motivated us to persevere and give our best.

To our research adviser, Ms. Rea Mie A. Omas-as, thank you for your invaluable guidance, encouragement, and dedication, which helped us overcome difficulties and complete this study successfully.

To our friends and classmates, your support, understanding, and companionship made this research journey meaningful and memorable.

This work would not have been possible without all of you.



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CHAPTER 1

INTRODUCTION

1.1 Project Context

Efficient and accessible civil registration systems are essential for accurately recording vital events like births, deaths, and marriages. According to the United Nations, strong civil registration and vital statistics (CRVS) systems are key for good governance, public health, and sustainable development [1]. However, paper-based methods often cause delays and limited accessibility, making it difficult for individuals to verify their legal identity and access essential services [2]. As governments transition to digital platforms, more countries recognize the value of e-government tools in enhancing transparency, accuracy, and citizen services.

The COVID-19 pandemic revealed both the strengths and weaknesses of global e-governance systems. Digital platforms definitely made things easier to access and more transparent. However, honestly, many government offices are still stuck in the Stone Age with their clunky systems and tech that belongs in a museum[3]. Several issues, including slow and inefficient paper-based processes, lengthy waiting times, manual form errors, and limited accessibility for citizens in rural areas, hinder the acquisition of Civil Registry Documents. These problems result in low user satisfaction, wasted time and resources, and contribute to environmental concerns due to high paper consumption and waste generation[4]. Both studies underline the urgent need for effective and transparent digital governance. While Villena et al.[4] Focus on local digitalization efforts in the Philippines, emphasizing efficiency and environmental sustainability, as noted by Naqvi et al.[3] Expand the perspective to global post-pandemic challenges, stressing collaboration, data design, and cloud-based adaptability. Together, they highlight a continuous progression of e-governance from basic digitization of services to developing robust, integrated, and crisis-ready governance systems. Tapangan [5] highlighted how digitalizing civil registration in the Philippines

significantly aided them in managing their operations during the pandemic. Villena et al.[4] essentially stated that making this information available online makes it significantly easier for everyone, especially those living in remote areas. Both studies highlight old problems: shaky infrastructure, wonky resource management, and inconsistent rules enforcement. So yeah, good ideas, but plenty of roadblocks.

On the international scale, Angin et al.[6] studied how Indonesia transformed its population and civil registration services during the COVID-19 pandemic by introducing the SILAO system. This system aimed to make registration contactless and efficient, but the study found several problems. Many staff and citizens struggled with the system due to its difficult navigation. The SILAO platform also required a strong internet connection, posing a challenge for users in remote areas. Some steps still had to be done in person at registration offices. Limited digital skills among employees and the public also slowed adoption. While SILAO was a step forward, it did not fully deliver an accessible and smooth digital service for everyone.

At the local level, Celeste et al. [7] assessed how extensively citizens in a municipality in Negros Occidental registered simple events such as births, marriages, and deaths. The findings indicated that birth and marriage registrations were typically well-managed, but death registrations were often not attended to, suggesting a lack of perceived value among the population. The study highlighted some challenges that rendered registration less effective, including unclear requirements, far-off office locations, and the absence of directions, particularly for rural and poor populations. Although the authors called for electronic systems to enhance the Civil Registry and Vital Statistics (CVRS), they failed to investigate how digital tools would simplify the process. Individuals continued to register in person with paper forms, with no website or notifications. These problems indicate a demand for solutions that are simple to use digitally. For this, the present study proposes a paperless registration system with email notifications and an

easy interface to make it more accessible, efficient, and responsive to community needs.

The present study proposes a digital system with real-time tracking, online appointment scheduling, and automated reports to address these persistent challenges. The system will be developed based on user needs, stakeholder feedback, and repeated testing to ensure it is easy to use and accurate. Key features will include role-based access, secure data backup, and notifications by email to keep users informed and ensure transparency. The system aims to reduce paperwork, avoid duplicate work, and improve accountability in the Civil Registrar's Office by moving document submission, checking, and storage online.

This study is critical because it helps advance local e-governance and digital public services. The system provides a lasting solution to enhance legal identity management and supports the Philippine government's initiative toward paperless, citizen-centric transactions. Kumar et al. [13] noted that digital tools in public offices build trust by making processes more transparent and efficient, while Vicente et al. [8] found that digital platforms make citizens happier by speeding up access and cutting wait times. This study addresses problems in the current manual process and shows how digital changes can improve local governance and encourage other towns to try similar solutions.

The new Online Transaction System for the Civil Registrar Office of Buenavista, Agusan del Norte, aims to improve how the municipality handles civil registration. Currently, the office uses paper forms for birth, marriage, and death certificates, which often causes slow service, lost or repeated records, and long waits for citizens. These problems delay document release and make it hard for staff and clients to track requests. The new system lets users submit documents online, check their application status, and access records in a safer and more organized way. Compared to Indonesia's SILAO system, which lacks user-friendly features such as automated email notifications [6], and the conventional paper-based registration procedure previously used in Negros Occidental

[7], this new system aims to simplify life. The proposed system provides users with automatic email notifications and features a user-friendly interface accessible to users with limited computer experience. It is a significant upgrade from the previous manual handling of records, as it saves people from spending excessive time traveling back and forth to multiple offices to obtain documentation. The digital method also reduces employees' workload by avoiding excessive paperwork and data management. Moreover, it also enables local authorities to obtain accurate and real-time information, replacing outdated and unorganized files. In sum, adopting this system significantly improves the efficiency, transparency, and usability of civil registration procedures in Buenavista.

REVIEW OF RELATED LITERATURE

This chapter outlines the relevant literature supporting the researchers' study of the Online Transaction System for the Civil Registrar's Office in Buenavista, Agusan del Norte. It highlights the problems in traditional registration methods, the benefits of computerized substitutes, and the need for user-centric design principles. Various studies, articles, journals, and other web-based papers discuss computerization in civil registration centers.

Government Civil Registration System

According to Naqvi et al.[3], The COVID-19 pandemic exposed both the advantages and the disadvantages of the world's e-governance systems. Since online governance enhanced transparency, access, and civil involvement, the organizations responsible for these systems poorly set them up to meet the sudden surge in demand. The authors highlighted three important factors for resilience: collaboration across government sectors, strong data management, and cloud-based services. Villena et al.[4], The transition to paperless e-government in the Philippines responds to citizens' requirements for faster, more transparent, and more accessible civil registration services. Their results were that online channels assisted in viewing such records as birth, marriage, and death certificates. Nonetheless, some deterrents remain, including low digital literacy for some segments of society, limited access in some regions, and common mistrust of online systems. It hinders adoption and creates gaps in inclusiveness.

Vicente et al.[8], emphasized that the digitalization of Peru's civil registration system (RENIEC) brought mixed results. According to the study, digital birth and divorce certificates typically raised user satisfaction. In contrast, digital death certificates frequently led to dissatisfaction among senior users who were not very at ease with the digital procedures. It suggests that even though digitalization simplifies matters for services and restricts paperwork, it is still a function of how

appropriate the system is to individual needs and abilities at different ages. The following studies illustrate the virtues and vices of taking services for civil records and governance to the online sphere. As Naqvi et al.[1] As mentioned, there should be firm and flexible systems. On the other hand, Santos et al. were concerned with citizens' preparedness and infrastructure support. Additionally, Vicente et al.[8] highlighted the roles of service design and population components in user satisfaction. In total, these thought processes mean that digitalized governance is not merely about digitizing service delivery, but about making sure that the systems are resilient, accessible, and responsive to the extensive needs of citizens. In our case study, successful e-government must integrate contemporary technology with infrastructure investment, promote citizens' awareness, and plan services with user diversity in mind.

Digital Transformation of Civil Registry Systems

According to Tapangan[5], the City Civil Registry Office in Cagayan de Oro, Philippines, uses digital technologies. Tapangan[5] further mentioned that these digital technologies help to enhance the efficiency and effectiveness of its operations, like the City Civil Registry Information System (CCRIS), digitized records, central database, and online applications. They replace the old manual ways, making records more accurate. Tapangan[5] also says this is more reliable. But Angin et al.[6] argue that using new technology after crises is hard. They require sturdy equipment, adequate infrastructure, trained staff, and efficient processes. Interestingly, Ndiyun et al.[2] argue that sustainability depends on addressing access and capacity gaps. For the current investigation, these lessons emphasize that digital systems must be sustainable, inclusive, and citizen-centric so that civil registration can support long-term development aspirations and unforeseeable challenges. More resilient. Those initiatives solved ubiquitous issues like time-consuming manual filing, disaster damage vulnerability, and extended office queuing. They also improved accuracy,

reduced costs, and made services more accessible for citizens, especially during the COVID-19 pandemic.

According to Lestari et al.[9], Civil registration and vital statistics (CRVS) systems are the most reliable data source for monitoring health-related Sustainable Development Goals (SDGs), particularly when linked with national identification systems. Researchers could link records generated from civil registration, health, and migration data if each citizen could be assigned a unique identification number at birth. It will provide more reliable data for governments to use for planning and resource allocation, making it easier for citizens to access many services, including healthcare, education, and social protection.

According to Shoewu et al.[10], Nigeria's birth/death system uses manual record generation, storage, and retrieval methods. That has resulted in error-prone, inefficient processes and the tendency to lose vital data, thus rendering population data and planning ineffective. The study authors created an online system to automate record-keeping, eliminate paperwork, and build a more robust information base for decision-making. This research highlighted that digital solutions are key to preserving rights, delivering services, and contributing to demographic planning. Researchers have to designed an online system that would maintain records in automatic mode, weeding out the paperwork and at the same time more accurate database for making any decision. The study emphasized that a digital system is crucial to safeguarding people's rights, adding value to services, and demographic planning.

According to Angin et al.[6], The COVID-19 pandemic pushed the digital transformation of public services in Indonesia, especially in civil registration. The Situbondo Civil Registry Office has introduced the SILAO online system, enabling citizens to apply for ID cards, family cards, and certificates from their homes. However, internet and staff knowledge issues still result in an

imperfect system. Ndiyun et al.[2] Cameroon made a significant change in 2024 by digitizing its civil registration system. Changes included electronic declaration and a database, making certificates electronic and improving access and transparency. This improved service delivery; however, there are still issues with weak internet access, low digital literacy, and inadequate infrastructure. The authors emphasized that policymakers and educators must address such barriers to achieve long-term effectiveness through improved policies and citizen education. They indicate that online conversion of civil registration can highly enhance efficiency, transparency, resilience, and service access. However, the correct support for systems depends on the infrastructure, inclusive policy, and citizens' preparedness.

Assessment of Civil Registration Awareness

According to Celeste et al.[7], Civil registration in the Philippines shows that most residents know and comply with registration requirements, especially for births and marriages. However, death registration remains less consistent. Their study also found that women and people with higher levels of education tend to comply more. Even with these efforts, obstacles like the unavailability of valid identification cards, transportation costs, and mix-up with documentary requirements still hinder some citizens from registering. In a study by Fruchtman et al. [11], the COVID-19 pandemic revealed how essential Civil Registration and Vital Statistics (CRVS) systems are in supporting public health and legal identity protection. Their research emphasized that resilient CRVS systems, capable of continuing operations during crises, adapting to disruptions, and protecting human rights, are critical for effective governance. The authors introduced the framework for Resilient CRVS Systems, which identifies seven key capacities that enable countries to prepare for, respond to, and recover from emergencies. Findings from case studies in countries such as Guatemala, Ecuador, and Fiji showed that nations with digitized, decentralized, and well-integrated CRVS systems could better maintain services and monitor mortality data during the

pandemic. The study recommends governments invest in digitalization, interoperability, decentralization, and inclusivity to strengthen CRVS resilience and ensure equitable access to civil registration services.

As defined by Mikkelsen et al.[12], CRVS systems play two important roles: giving legal identity through birth registration and providing vital fertility data to governments. Although global initiatives have focused on expanding birth registration towards attaining Sustainable Development Goal (SDG) 16.9, the research highlighted that analysts have not yet fully utilized the value of CRVS systems in gathering high-quality fertility data. If utilized effectively, good birth data can inform population projections and assist governments in planning for schools, clinics, and hospitals. Separately, the works establish that civil registration keeps improving but remains bedeviled by issues that take away from the effectiveness of civil registration as a complete entity. As noted by Celeste et al.[7], It is crucial to address the issues that individuals encounter, including cost and the requirement for official identification. Fruchtman et al. [11] emphasize that Governments must strengthen Civil Registration and Vital Statistics (CRVS) systems through digitalization, interoperability, and good governance to remain functional and resilient during emergencies such as the COVID-19 pandemic. This study's results indicate that a strong civil registration system includes solutions targeting citizens and institutional improvement, and maximizes the use of registration data to support country development.

Secure and Efficient Management of Vital Records and Population Data

Sudan Kumar Mulmi [13] states that the vital registration system in Nepal has been improving over the years, with an increase in the number of people registering births, deaths, marriages, divorces, and internal migrations. Nonetheless, the lack of awareness about the importance of registration remains a reason for many to delay it, while outdated laws also slow down the process. The study pointed out that raising public awareness and updating legal provisions are necessary

to make the system more reliable and efficient. According to Herlinda et al.[14], The COVID-19 pandemic encouraged digital services in Indonesia's population administration, such as the SIPENDUDUK online system in Pekanbaru. This platform made services more accessible, but faced problems like server breakdowns, limited processing capacity, and low public awareness. The SIPENDUDUK platform, on the one hand, has brought the population administration service closer to the public, allowing them to make submissions from home. However, the system still faces several fundamental implementation challenges, such as frequent server breakdowns and limited data processing capacity, while using SIPENDUDUK. The authors stressed that technology alone is not enough; The SIPENDUDUK must promote proper preparation, and educators must actively use public education to make digital systems effective.

According to Sumirat et al.[15], Indonesia introduced absolute liability statements (SPTJM) to help families from unregistered marriages get critical population documents like birth certificates. While this made access easier, the study raised concerns about legal certainty, especially regarding inheritance rights. The authors explained that although SPTJM helps in the short term, it cannot replace formal legal processes such as court confirmation of marriages. Sordo et al.[16], say that late registration of deaths makes it hard to track deaths during health emergencies like the COVID-19 pandemic. They developed a statistical model to obtain nearly all the weekly death counts from incomplete data. It provides less biased, timelier information and allows governments to respond better to crises.

Nalband et al.[17] Manual civil registration systems using paper-based systems are inefficient and suffer from many issues, such as record forgery, loss, and untimeliness. They developed a digital civil registration system using ASP.NET and SQL Server to overcome the challenges. The system performed the tasks of data input, document verification, and certificate generation, while at the same time, it enhanced security through encryption and QR codes. The various studies

highlight that civil registration has improved, but issues still need to be addressed. Sudan Kumar Mulmi [13] stated that we must update laws and educate about the civil registration process, and Herlinda et al.[14], stated that digital systems need technology and community engagement to work correctly. Sumirat et al.[15], pointed out the paradox between facilitating access to services and legally protecting these services. Sordo et al. [16] showed how new methodologies can improve the time-lag in data reporting. The main point was that Nalband et al.[17], demonstrated that electronic platforms could replace the traditional paper systems in civil registration, gaining better accuracy and security. The researchers' heads-up indicated that civil registration modernization must consider not only hardware improvements but also legal reforms, public consciousness, and access and dependability strategies.

Vital Statistics: A User-Friendly System for Efficiency and Sustainability

According to Vicente et al.[8], The move toward digital civil registration in Peru, managed by RENIEC, has improved accessibility and efficiency. Their study showed that digital birth and divorce registration platforms increased user satisfaction. In contrast, death registration had mixed results because older users found it challenging to adapt to online systems. In Indonesia, Anityasari et al.[18], looked at the transition of Surabaya's civil registration services into digital platforms like KLAMPID COVID. They found online systems sped up processes, reduced paper usage, and helped with health regulations during the pandemic. However, it also increased electricity and internet usage.

In a similar way, Herdianti et al.[19], worked on a mobile web-based system in Binjai City because registration took too long, and the line was always long. The new system made things smoother, with fewer errors, and users liked it a lot. The research shows that when applied in civil registration, digital technology speeds up the entire process and makes it user-friendly; therefore,

trust is built. They highlight that technology alone is not enough to solve the issues in civil registration. Decision-makers must consider practical barriers such as costs and access. The present study suggests that civil registration reforms combine technology with user-centered design and supportive policies to ensure that services are efficient but also inclusive and sustainable.

Web-Based Application System

According to Kaewsuwan et al. [28], creating a web-based registration system was to solve the problems with manually enrolling students at Pitchayabundit College in Thailand. Their research utilized a digital platform to streamline registration, record-keeping, and grade submission tasks. The system reduced the likelihood of mistakes, accelerated processing, and increased user satisfaction. The researchers said that using information technology in administrative work makes data more accurate, easier to find, and more efficient overall. It shows how going digital can make old ways of keeping records more up-to-date.

According to Celeste et al. [29], civil registration constitutes the legal documentation of significant events, including births, marriages, and deaths, as stipulated by Republic Act No. 3753, recognized as the Civil Registry Law of the Philippines. Their research examined the levels of awareness, compliance, and challenges encountered by residents of Negros Occidental municipalities concerning civil registration. The results showed that people were very aware of and followed the rules for registering births and marriages, but not so much for registering deaths. The study demonstrated that education, gender, lack of identification, and limited access to places influenced the results. It highlighted that increasing awareness of and access to services is crucial for encouraging more people to participate in government documentation processes.

According to Vijay Panwar [30], digital civil registration and legal identity systems are vital in promoting inclusivity and accountability in governance. The report argued that traditional paper-based registries are inadequate in addressing the needs of modern society, as they often exclude marginalized groups from legal recognition and essential public services. By shifting to digital civil registration and vital statistics (CRVS) systems, countries can enhance data accuracy, accessibility, and timeliness.

The study emphasized that integrating civil registration with identity management and other government systems will ensure everyone participates. It aligns with Sustainable Development Goal (SDG) 16.9, which promotes "legal identity for all by 2030." *The* reviewed studies collectively emphasize the transformative impact of digital systems in improving administrative processes, service accessibility, and data management. Kaewsuwan et al.[28] demonstrated how web-based applications streamline operations and reduce inefficiencies in institutional workflows. Similarly, Celeste et al. [29] highlighted the need for accessible and user-friendly systems to encourage citizen participation in civil documentation. Vijay Panwar [30] expanded this perspective by underscoring the importance of digital civil registration systems in ensuring inclusivity and accurate data management for effective governance. Together, these studies support the current research's aim to develop an online documentation requisition system that promotes efficiency, transparency, and accessibility for local citizens.



1.3 Objective of the Study

The researchers' study aims to design and develop an online transaction system for the Civil Registrar's Office of Buenavista, Agusan del Norte, to streamline civil registration processes and enhance service delivery for citizens.

Specifically, it aims to:

1. Digitalize and streamline the Local Civil Registry (LCR) transactions of the Municipality of Buenavista LGU, particularly focusing on birth, marriage, and death registration services to ensure faster and more accurate processing.
2. Develop an online system that allows applicants to upload necessary documents, track their application status in real time, and receive updates through email notifications.
3. Implement ethical standards in data management by implementing encryption or hashing techniques that ensure the confidentiality, integrity, and security of sensitive information, thereby protecting user privacy and preventing unauthorized access or data misuse.
4. Evaluate the developed system based on the ISO 25010 Software Quality Standards to assess its functionality, efficiency, and overall usability.

1.4 Scope and Limitations

This capstone project seeks to design and develop a web-based Online Transaction System for the Civil Registrar's Office of Buenavista to replace manual processes for civil registration services such as birth, marriage, and death registrations. The system is only for the bona fide residents of Buenavista who present proper identification. The users can submit applications, track their status, and receive notifications through the online system, while designated personnel from

the Civil Registrar shall handle transactions, evaluate applications, and generate reports through a standard administrator account.

The system will operate exclusively in the Civil Registrar's Office, but users can also access it through computers and cell phones. It supports online access to provide clients with convenient usage anytime and anywhere, as long as an internet connection exists. The system will not support online payment since the Treasurer's Office will process all payments separately.

1.5. Definition of Terms

Agile Methodology – a development approach used in creating the system, where features are built and improved through iterative and incremental cycles to ensure continuous refinement.

Civil Registry Documents – the records managed and processed by the system, such as birth, marriage, and death certificates, which are stored and retrieved from the Local Civil Registry database.

Data Hashing – a security feature of the system that converts sensitive information (such as passwords) into irreversible codes to protect user credentials.

Data Validation – an automated process within the system that checks the accuracy, completeness, and correctness of user-submitted information before final submission.

Notification System – a feature of the Online Application System that automatically sends updates and status alerts to users through email.

Online Application Portal – the web-based module of the system where users can apply for civil registry documents and submit all required information.

Real-Time Tracking – a function of the system that allows users to instantly view the current status of their document request as it changes.

Record Management – the process within the system responsible for systematically storing, organizing, updating, and displaying civil registry records in the database.

Role-Based Access Control (RBAC) – a security mechanism of the system that grants or restricts access to features depending on the user's assigned role (e.g., admin, staff, applicant).

CHAPTER 2

TECHNICAL BACKGROUND

Research Design

This study will use a Design and Development Research (DDR) approach, which involves creating and testing the Online Civil Registry System. This technology-based solution enhances record management, minimizes processing delays, and increases transparency within civil registration procedures.

Software Development Life Cycle



Figure 1. Agile Methodology

Figure 1 illustrates the use of the Agile Methodology Software Development Life Cycle in developing the Online Civil Registry System. Agile methodology uses short, repeated sprints to design, build, and test functional parts. This method allows for flexibility and encourages collaboration among LCR Staff instead of following a set order. The first step is to plan and gather requirements. Most important features, like user logging, usage tracking, and reporting. Each sprint includes the design, development, and testing of separate modules, making it easy to use

feedback to improve the system's efficiency. This Agile method ensures the system is user-friendly, flexible, and capable of handling the evolving needs of civil registration services.

2.1 System Architecture

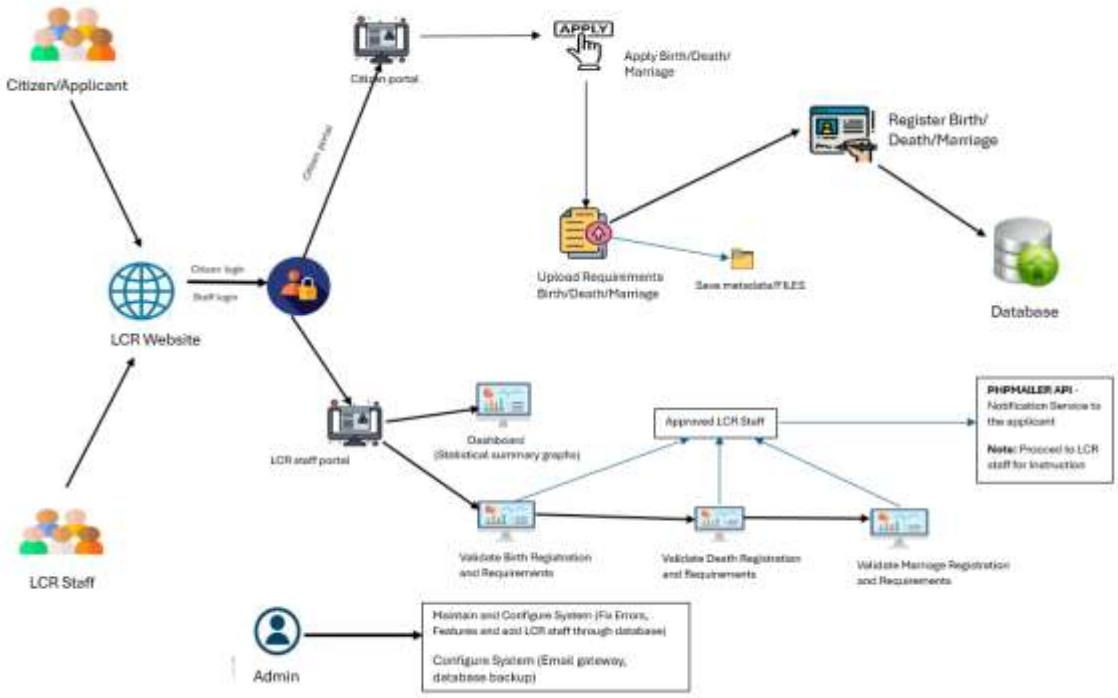


Figure 2: System Architecture

Figure 2 presents the three-tier architecture of the Online Transaction System, comprising the Client, Application, and Data Tiers. The Client Tier delivers the user interface, developed with HTML, CSS, and JavaScript, and supports access via web browsers. The Application Tier executes business logic using PHP, manages authentication and role-based access control, handles file storage and notifications, and produces reports including graphical representations and PDF exports.

2.2 Conceptual Diagram

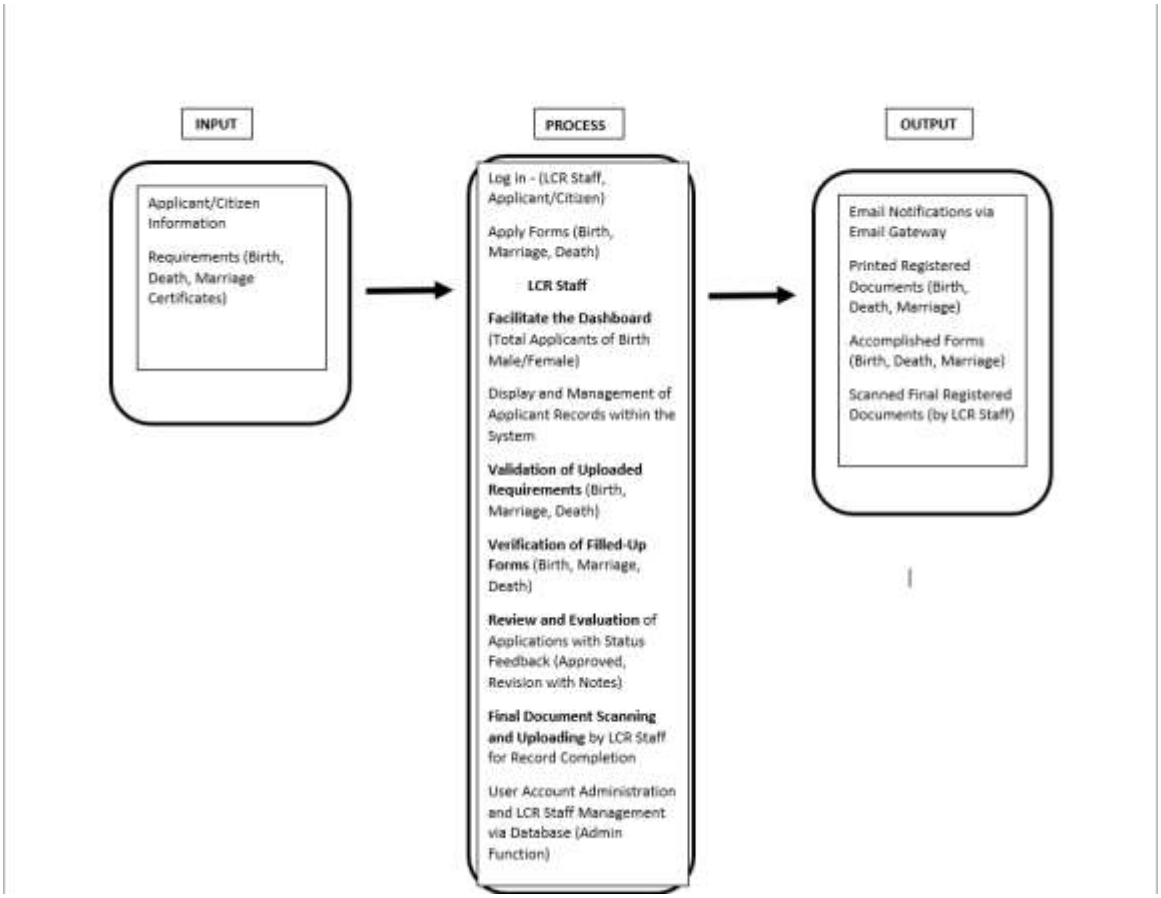


Figure 3: Conceptual Diagram

Figure 3 illustrates the conceptual diagram of the Local Civil Registry Office's (LCR) Online Transaction system. In addition to creating profiles and logging in, applicants can upload necessary documents, register significant life events such as births, marriages, and deaths, and track the progress of their applications. Applicants receive email notifications or text messages to keep them updated on the status of their applications. LCR staff review and check applications, review the uploaded documents, and decide whether to approve, deny, or ask for changes. Staff members also write reports that examine the births that staff have registered. The administrator or

programmer keeps the system running by setting up features, fixing errors, and running the system.

2.3 Use Case Diagram

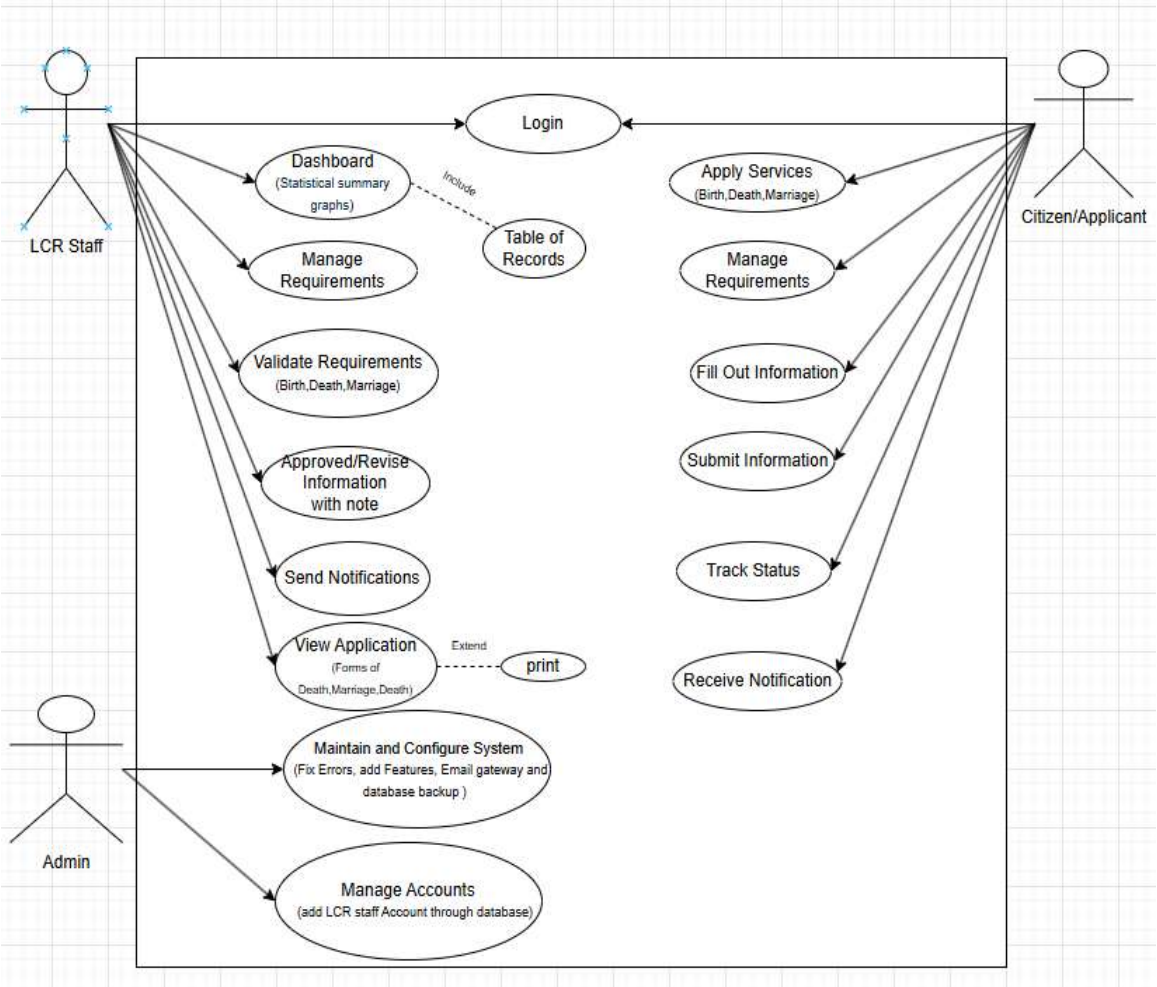


Figure 4: Use Case Diagram

Figure 4 shows the use case diagram of The Local Civil Registry Office's (LCR) Online Transaction System. In addition to creating profiles and logging in, applicants can upload necessary documents, register important events like births, marriages, and deaths, and track the progress of their applications. The progress will notify applicants via Email. LCR staff evaluate applications, verify supporting documents, and decide on approval, rejection, or the need for revisions. System

stability relies on feature configuration, technical issue resolution, and operational management by the admin or programmer.

2.4 Activity Diagram

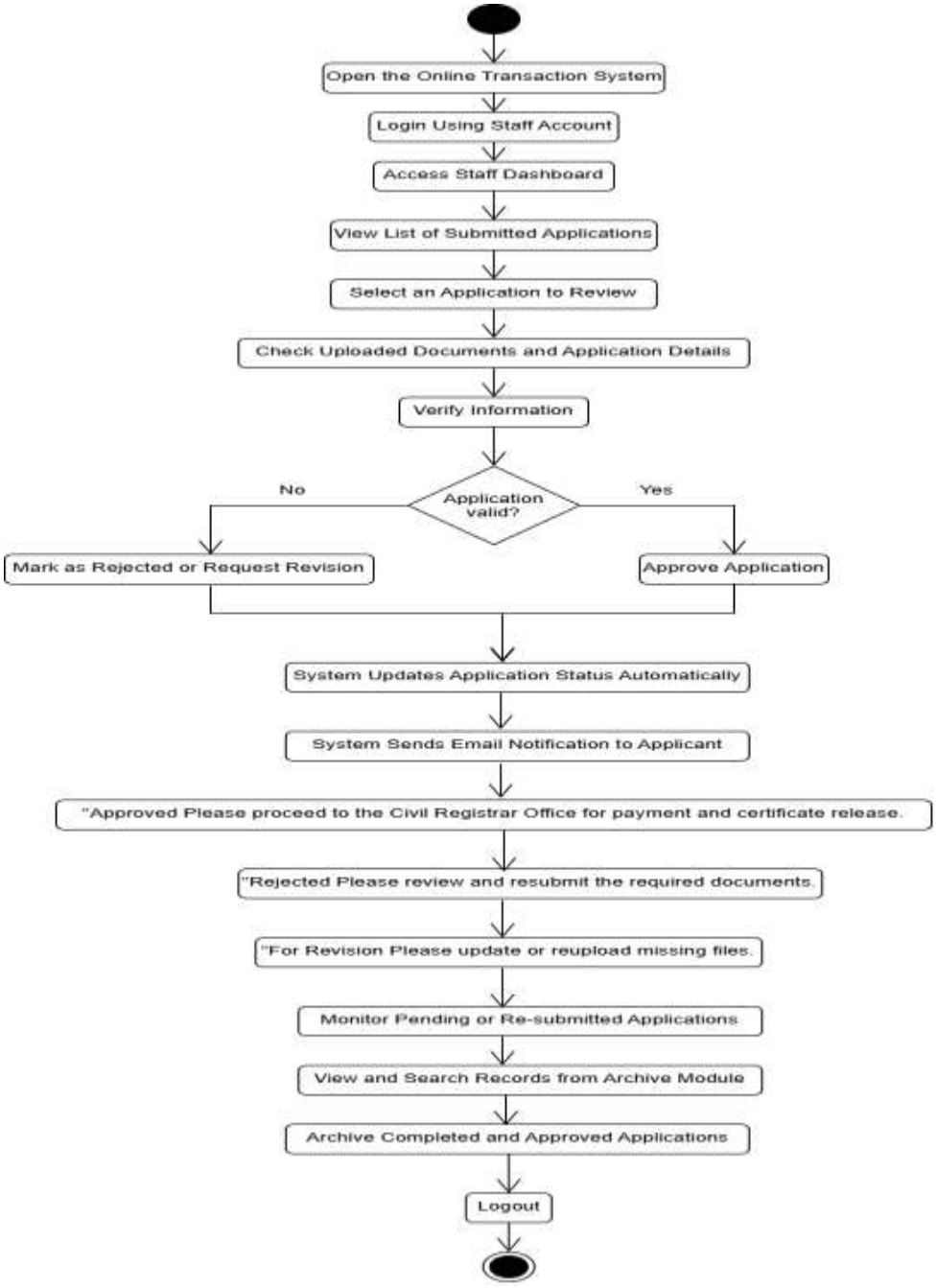


Figure 5: Activity Diagram LCR Staff

Figure 5 illustrates the activity diagram workflow of the Local Civil Registry (LCR) Staff as they manage and handle online civil registration applications. The process starts when staff members log in with their staff account and open the online civil transaction system. Once on the staff dashboard, they can view submitted applications and select one to review. The staff reviews the uploaded documents and application information to ensure everything is correct. The staff may reject the application or request a revision if it is invalid. If valid, the staff approves it. The system will automatically update the application status and send an email notification to the applicant.

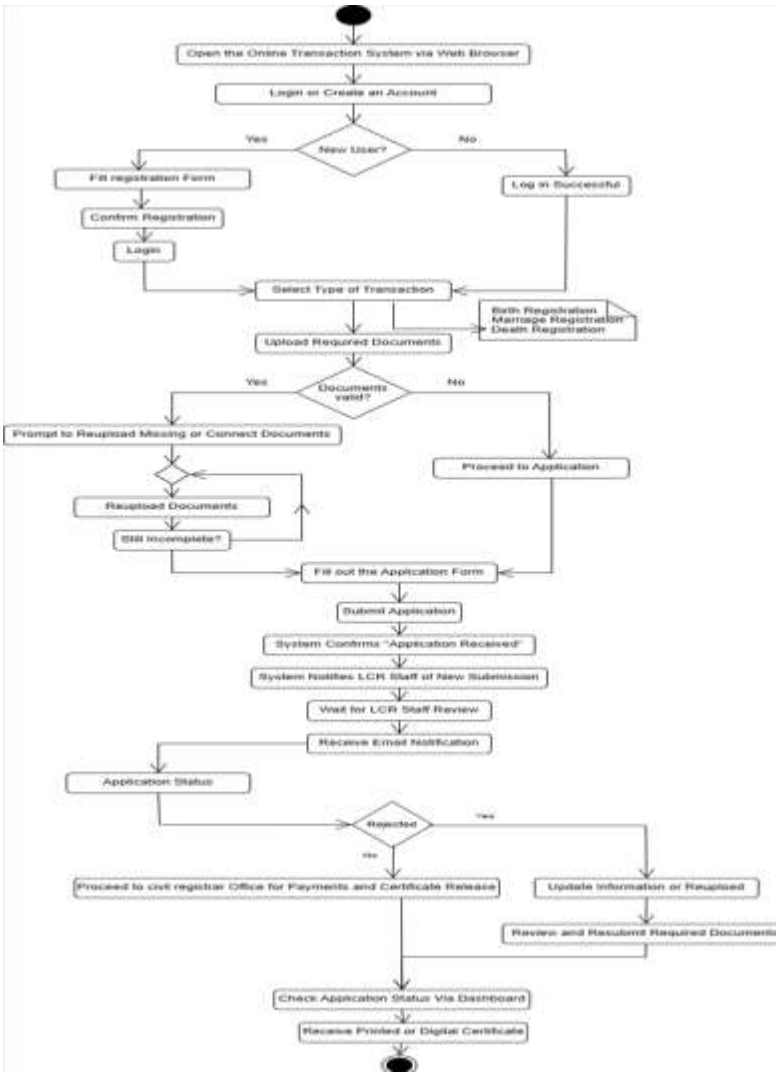


Figure 6: Activity Diagram Citizen/Applicant Staff

Figure 6 illustrates the activity diagram flow for an applicant or citizen using the online civil registration system. The process starts with the user opening the online transaction system through a web browser and logging in or registering for an account. If the user is new, they must complete the registration form and validate their registration through email before logging in. After logging in, the applicant selects the type of transaction, i.e., Birth Registration, Marriage Registration, or Death Registration. The system then asks the user to upload documents subsequently. If the documents are incomplete or incorrect, the system asks the applicant to

reupload or edit the documents accordingly. Once all the documents are accurate, the applicant can submit the application form.

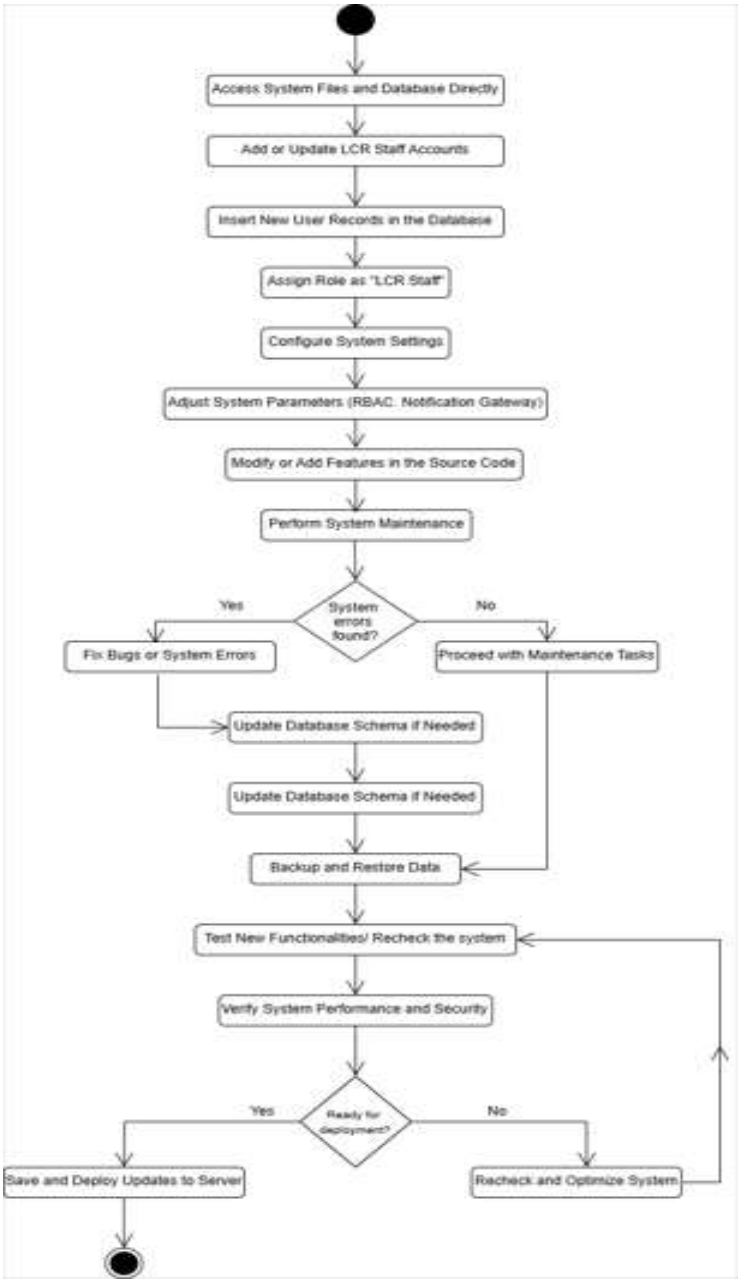


Figure 7: Activity Diagram Admin/Programmer

Figure 7 shows the Sequence Diagram of a Programmer or Admin, outlining the technical tasks involved in maintaining, updating, and managing the Local Civil Registry system. The admin

can create or update LCR staff accounts, add new user records to the database, and assign roles such as "Staff associated with the LCR." After that, the admin sets up the system's settings and updates essential settings, such as Role-Based Access Control (RBAC) and the Notification Gateway, to ensure the system operates smoothly and securely.

2.5 Sequence Diagram

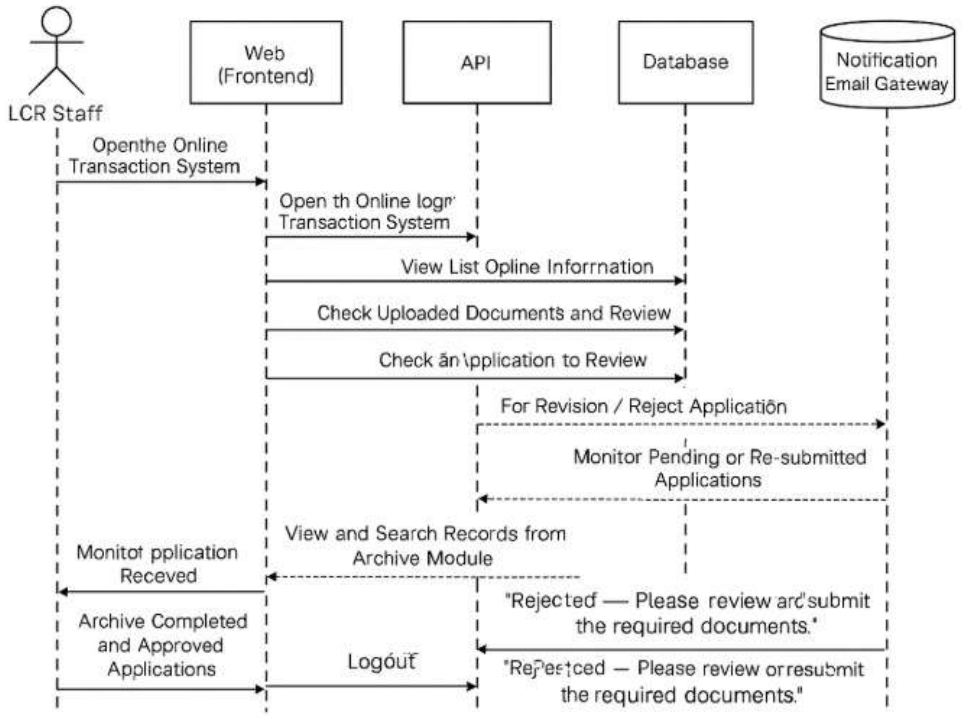


Figure 8: LCR Staff Sequence Diagram

Figure 8 illustrates how the LCR staff members work within the system. Staff members register or authenticate via the web interface. Upon authentication, they access a dashboard that presents the total number of registered births categorized by gender. Staff review applications in the processing queue and perform actions, including approval, rejection, or requests for revision. Following each decision, the system communicates the application status to the applicant. This workflow enhances verification, transparency, and accountability in civil registration management.

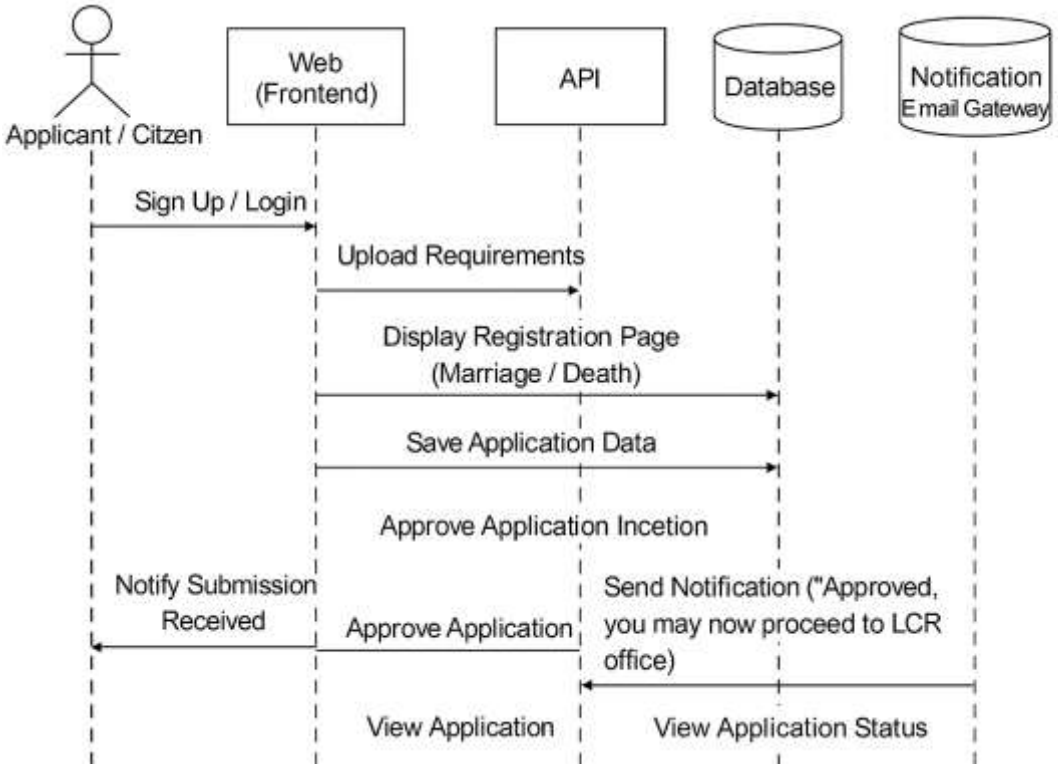


Figure 9: Applicant/Citizen Sequence *Diagram*

Figure 9 shows the steps an applicant or citizen follows when using the system. After signing up or logging in through the web interface, users can register or fill out applications for birth, marriage, or death certificates. Depending on the request, the system provides the specific requirements (early or at-early registration for birth or death certificates, or marriage requirements). Upon submission of the application, the system generates an automated confirmation. Following review and approval by administrative personnel, an email containing instructions for visiting the LCR office and completing the process is sent to the applicant. This procedure enhances usability, increases accessibility, and enables efficient application status tracking.

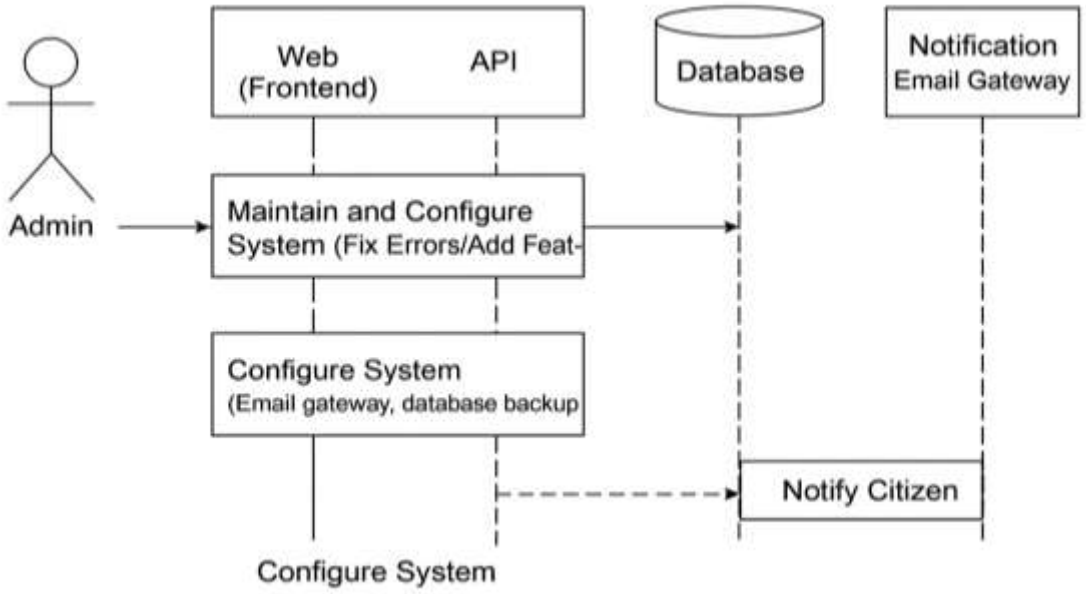


Figure 10: Admin/Programmer Sequence *Diagram*

Figure 10 illustrates the actions performed by administrators or programmers within the system. They fix bugs, update software, and add new features to keep things running smoothly. They also set up essential components, such as the email gateway and database backup, to ensure secure communication and data protection. Administrators help keep the system running smoothly and safely by working with the API and database.

2.6 ERD (Database Design)

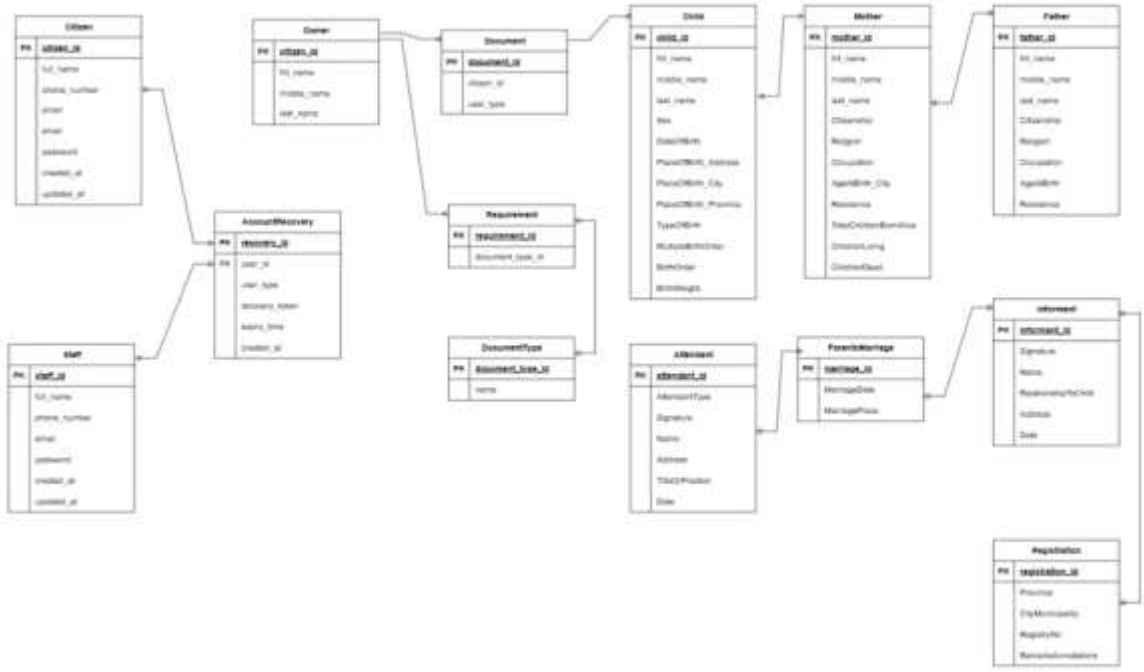


Figure 11: Entity Relationship Diagram (ERD) Diagram

Figure 11 illustrates the Entity Relationship Diagram (ERD) that the Local Civil Registry System organizes its data. It illustrates how the database entities support the handling of civil registration records.

2.7 User Interface Design (Prototype)



Figure 12: Landing Page Dashboard

Figure 12 displays the online services interface for the Local Civil Registry Office of Buenavista. The page features options for registering births, deaths, marriages, etc.



Figure 13: Citizen/Applicant Login Page

Figure 13 shows the login interface for the Local Civil Registry Office of Buenavista. It allows citizens or applicants to sign in using their email and password to access online services. Additional options include administrator login, account creation for new users, and password recovery for existing users.



Figure 14: LCR Staff Login Page

Figure 14 illustrates the login interface for the Local Civil Registry Office of Buenavista. LCR Staff can sign in using their email and password to access online services.



Figure 15: Account Creation Page

Figure 15 shows the form that new users must fill out to register for an account at the Buenavista Local Civil Registry Office. To make an account, applicants must give their full name, email address, password, and phone number (optional). This feature enables users to access the system's online services safely.



Figure 16: Citizen Portal of the Local Civil Registry System

Figure 16 shows the users' landing page after logging in to the Buenavista Local Civil Registry Office system. The interface provides users with direct links to important services, such as

registering a marriage, birth, or death, and addresses them by name. Users can register for each service on time or late and apply for the necessary documents through the system.



Figure 17: Birth Registration Requirements Page

Figure 17 presents the requirements section for death registration at the Local Civil Registry Office of Buenavista. Applicants can select either early/on-time registration or delayed/late registration. The page also provides a detailed list of documentary requirements.



Figure 18: Death Registration Requirements Page

Figure 18 illustrates the requirements section for birth registration at the Local Civil Registry Office of Buenavista. Applicants can select either early/on-time registration or delayed/late registration. The page also provides a detailed list of documentary requirements.



Figure 19: Marriage Registration Requirements Page

Figure 19 shows the requirements for getting married at the Local Civil Registry Office of Buenavista. The page also provides a comprehensive list of the necessary documents.

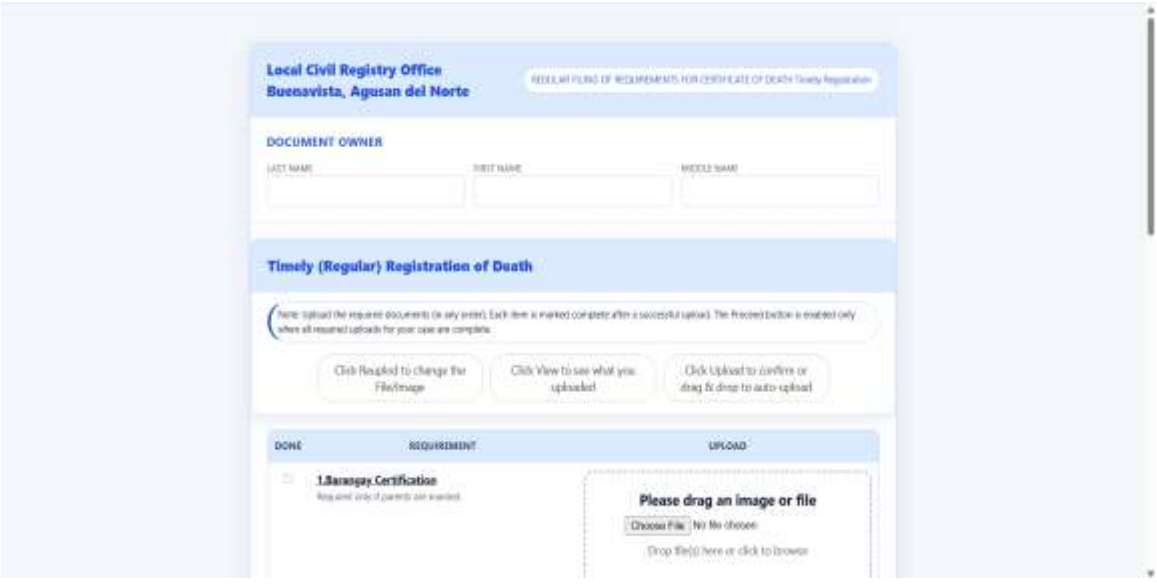


Figure 20: Registration Requirements Page

Figure 20 shows the online form used to register births, marriages, and deaths at the Buenavista Local Civil Registry Office. Applicants can upload the required files and input the details of the document owner. To ensure that all requirements, such as the barangay certification and other supporting documents, are submitted before proceeding, the form allows users to reupload, check, and confirm their documents.

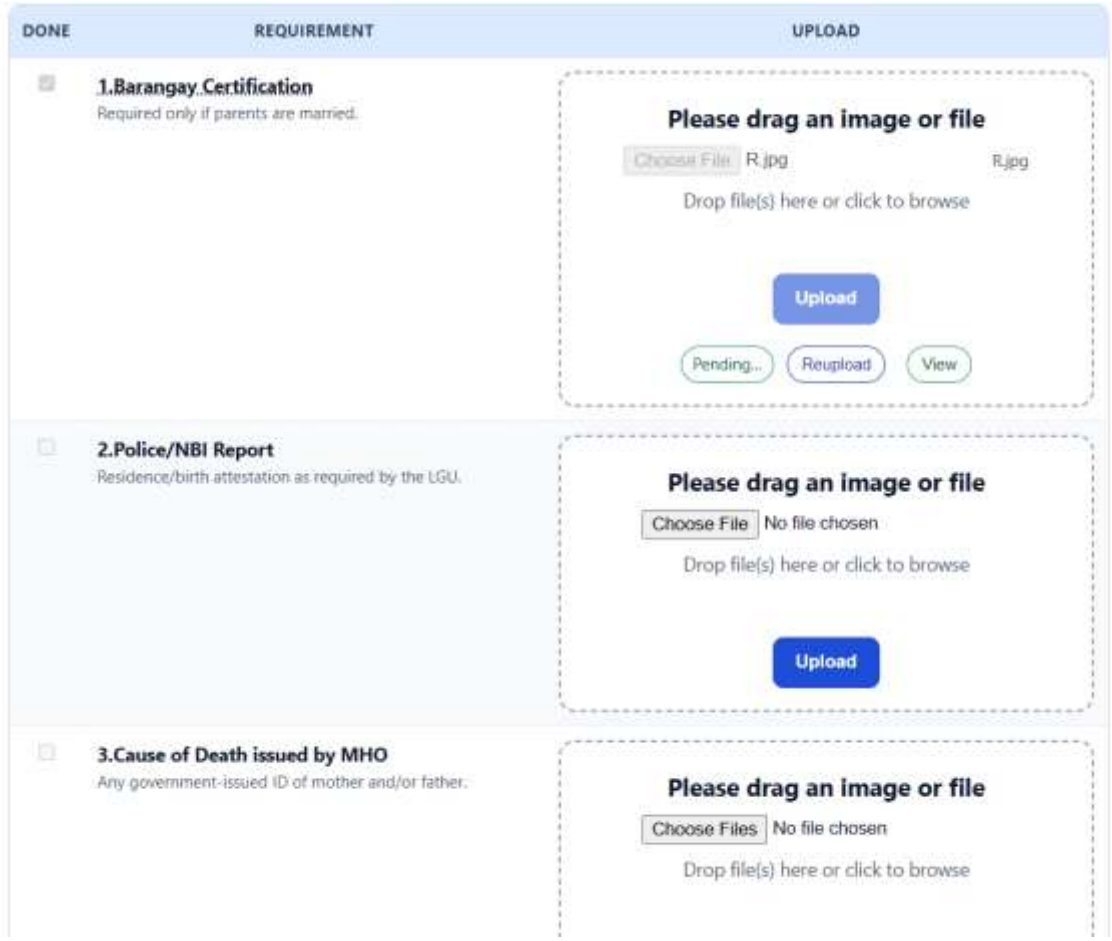


Figure 21: File Upload Section Registration Requirements

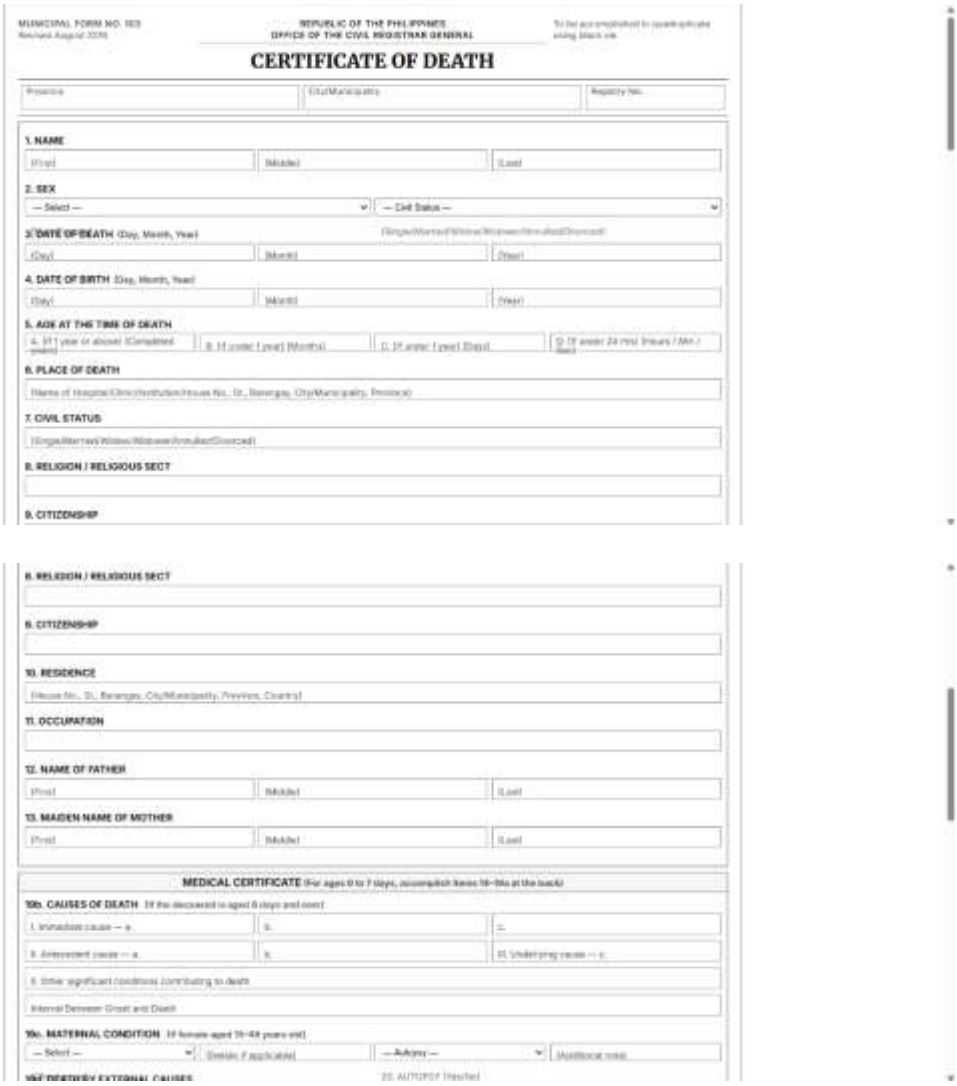
Figure 21 displays the document upload interface for the death, birth, and marriage registration process at the Local Civil Registry Office of Buenavista. It allows users to upload the

necessary documentary requirements. Each requirement section includes options to drag and drop files, upload, reupload, or view the status of the uploaded requirements, such as pending/approved, along with the submitted documents. Once a submission is either approved or marked for requirements revision, the applicant is automatically notified through the system and receives an email notification. The email clearly indicates the status of the submission, whether it has been approved or requires further revision and provides any remarks or instructions from the staff.



Figure 22: Certificate of Live Birth Form

Figure 22 illustrates the Certificate of Live Birth registration form under the Civil Registry System. The LCR staff (Local Civil Registrar staff) review the form after the applicant completes it. The staff can check all the input details before finalizing the record. At the top-right corner of the page, there are two options: Print the form or close the window, allowing staff to print a hard copy of the applicant's birth certificate for official use.



CERTIFICATE OF DEATH

Figure 23: Certificate of Death Form

Figure 24 shows the digitized version of the Certificate of Death (Municipal Form No. 103, Revised August 2016) employed by the Local Civil Registry Office of Buenavista. This digital format

ensures adherence to the standard document requirements of the Philippine Statistics Authority (PSA) for correct and systematic death registration.

MUNICIPAL FORM 90 (FORM NO. 2)
Revised January 2007

REPUBLIC OF THE PHILIPPINES
OFFICE OF THE CIVIL REGISTRAR GENERAL

To be accomplished in quadruplicate using black ink

APPLICATION FOR MARRIAGE LICENSE

Province	City/Municipality	Received by	Date of Receipt	Registry No.
			dd/mm/yyyy	

Marriage License No.	Date of Issuance of Marriage License
	dd/mm/yyyy

GROOM

THE CIVIL REGISTRAR

Sir/Madam:

May I apply for a license to contract marriage with _____ and to this effect, being duly sworn, I hereby depose and say that I have all the necessary qualifications and none of the legal disqualifications to contract the said marriage, and that the following data are true and correct to the best of my knowledge and information:

1. Name of Applicant
(First) (Middle) (Last)

2. Date of Birth/ Age
(Day) (Month) (Year) (Age)

3. Place of Birth
(City/Municipality) (Province) (Country)

4. Sex/ Citizenship
— Select — (Male/Female) (Citizenship)

5. Residence
(House No., St., Barangay, City/Municipality, Province, Country)

6. Religion/ Religious Sect

7. Civil Status

8. IF PREVIOUSLY MARRIED: How was it dissolved?

9. Place where dissolved
(City/Municipality) (Province) (Country)

10. Date when dissolved
(Day) (Month) (Year)

11. Degree of relationship of contracting parties

12. Name of Father
(First) (Middle) (Last)

13. Citizenship

14. Residence
(House No., St., Barangay, City/Municipality, Province, Country)

15. Maiden Name of Mother
(First) (Middle) (Last)

16. Citizenship

17. Residence
(House No., St., Barangay, City/Municipality, Province, Country)

18. Persons who gave consent or advice

19. Relationship

BRIDE

THE CIVIL REGISTRAR

Sir/Madam:

May I apply for a license to contract marriage with _____ and to this effect, being duly sworn, I hereby depose and say that I have all the necessary qualifications and none of the legal disqualifications to contract the said marriage, and that the following data are true and correct to the best of my knowledge and information:

1. Name of Applicant
(First) (Middle) (Last)

2. Date of Birth/ Age
(Day) (Month) (Year) (Age)

3. Place of Birth
(City/Municipality) (Province) (Country)

4. Sex/ Citizenship
— Select — (Male/Female) (Citizenship)

5. Residence
(House No., St., Barangay, City/Municipality, Province, Country)

6. Religion/ Religious Sect

7. Civil Status

8. IF PREVIOUSLY MARRIED: How was it dissolved?

9. Place where dissolved
(City/Municipality) (Province) (Country)

10. Date when dissolved
(Day) (Month) (Year)

11. Degree of relationship of contracting parties

12. Name of Father
(First) (Middle) (Last)

13. Citizenship

14. Residence
(House No., St., Barangay, City/Municipality, Province, Country)

15. Maiden Name of Mother
(First) (Middle) (Last)

16. Citizenship

17. Residence
(House No., St., Barangay, City/Municipality, Province, Country)

18. Persons who gave consent or advice

19. Relationship

Figure 24: Application for Marriage License Form

Figure 24 illustrates the digital version of the Application for a marriage license used in the Local Civil Registry Office of Buenavista. The organizer divides the form into two sections, one for the groom and one for the bride. Each blank requests personal information, including name, date and place of birth, age, sex, citizenship, and address. It also has areas for the Civil Registrar to verify that applicants meet the requirements for marriage in the Philippines and comply with the law.



Figure 25: Civil Registry Staff Dashboard

Figure 25 illustrates the dashboard staff use at the Buenavista Local Civil Registry Office. It helps administrators track and check records of marriages, deaths, and births. It has menus for

getting around, an overview of the system and its records, and tools for keeping an eye on applications, checking documents, and ensuring record keeping.



Figure 26: Statistics Dashboard

Figure 26 shows the statistics dashboard for the Buenavista Local Civil Registry Office. It shows the number of birth, death, and marriage records, along with a brief description. Staff can easily keep track of trends and compare registration data with the help of visual charts, such as a doughnut chart based on gender and a monthly bar chart.



Figure 27:Timely Birth Records Marital Table

Figure 27 displays the birth records marital section of the Local Civil Registry System of Buenavista. The interface includes action buttons for editing or deleting records and a search bar for quick retrieval. This feature allows authorized personnel to efficiently manage, update, and organize birth registration data within the system.

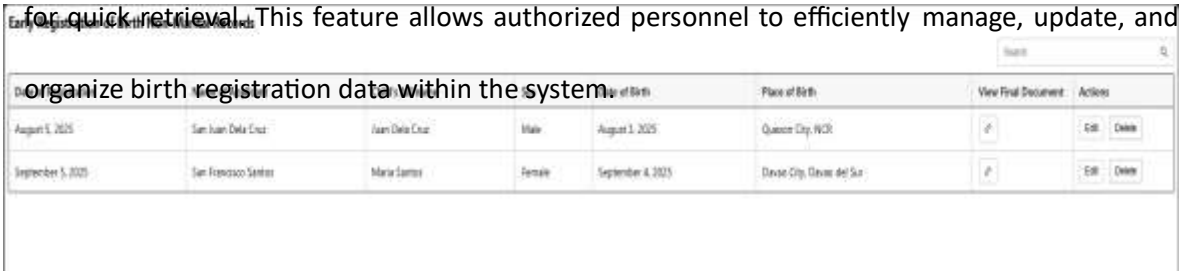


Figure 27.1: Timely Birth Records Non-Marital Table

Figure 27.1 displays the birth records non-marital section of the Local Civil Registry System of Buenavista. The interface includes action buttons for editing or deleting records and a search bar for quick retrieval. This feature allows authorized personnel to efficiently manage, update, and organize birth registration data within the system.



Figure 27.2: Birth Records Delay Table

Figure 27.2 displays the birth records delay section of the Local Civil Registry System of Buenavista. The interface includes action buttons for editing or deleting records and a search bar

for quick retrieval. This feature allows authorized personnel to efficiently manage, update, and organize birth registration data within the system.



Figure 28:Timely Death Records Table

Figure 28 shows the early death records table in the Local Civil Registry System of Buenavista, which keeps track of who has died. The LCR Staff can change or delete records with the buttons, and the search bar makes it easy to find records. This feature enables authorized users to easily add, modify, and organize death registration data within the system.



Figure 28.1:Delay Death Records Table

Figure 28.1 shows the delay death records table in the Local Civil Registry System of Buenavista, which keeps track of who has died. The LCR Staff can change or delete records with the buttons, and the search bar makes it easy to find records. This feature enables authorized users to easily add, modify, and organize death registration data within the system.



Figure 29: Marriage Records Table

Figure 29 shows that the Local Civil Registry System of Buenavista has a section called Birth Record. The buttons on the interface allow the LCR Staff to change or delete records, and the search bar makes it easy to find them. This feature enables authorized users to access, modify, and organize birth registration information within the system.

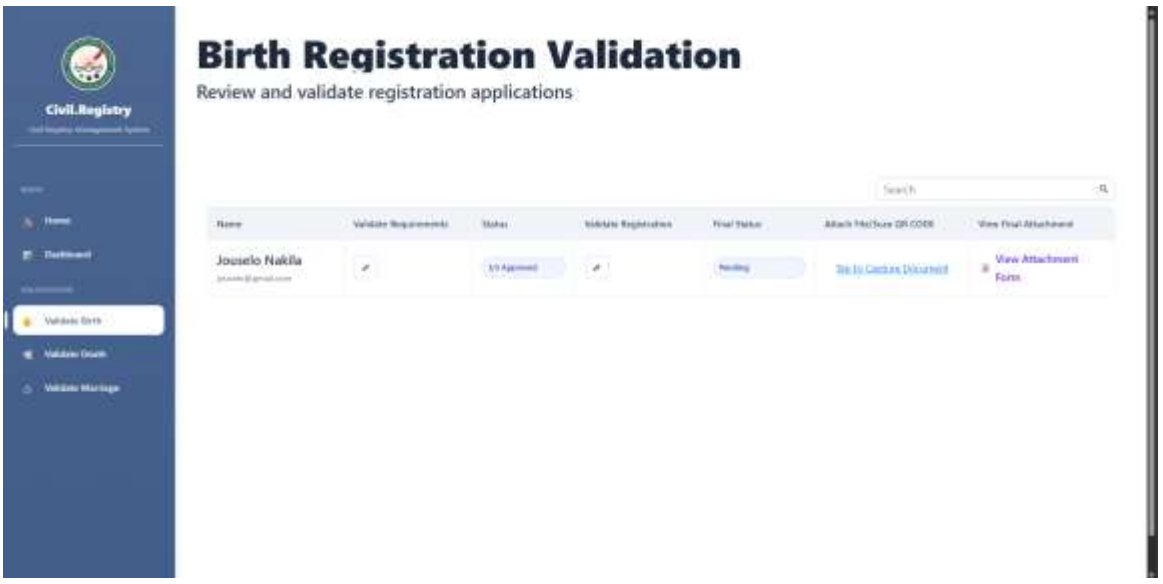


Figure 30: Birth Registration Validation Interface

Figure 30 shows the Birth Registration Validation part of the Civil Registry Management System. It allows authorized individuals to review and approve birth registration applications. The interface displays the applicant's name and lets them view the requirements and their registration status. The system displays the current status (Pending) and allows you to review the registration form again.

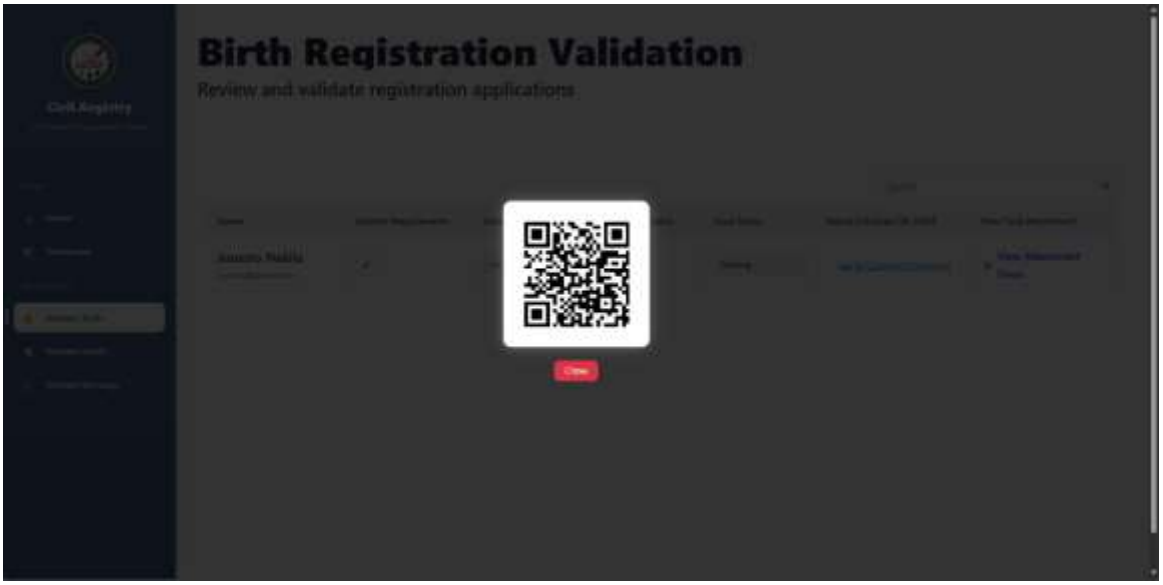


Figure 30.1: Birth Registration QR code

Figure 30.1 illustrates the QR code display interface of the Birth Registration Validation module in the Civil Registry Management System. The system generates a unique QR code for each birth registration record. When this QR code is scanned using the camera, it automatically redirects the user to the CS Scanner page or module, allowing staff to verify and capture the corresponding registration document. A “Close” button is provided below the QR code to return to the main validation interface.

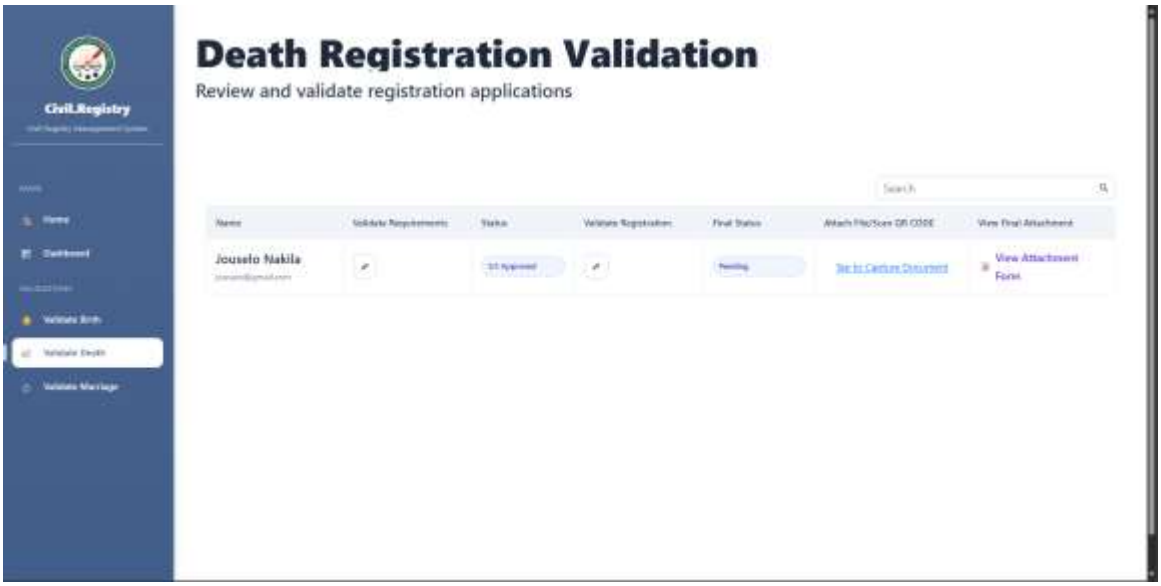


Figure 31: Death Registration Validation Interface

Figure 31 shows the Civil Registry Management System Death Registration Validation module. It lets people who have permission check and approve applications for death registration. The interface displays the applicant's name and provides options for checking their requirements and registration status. The system displays the current status (Pending) and allows you to review the registration form again.

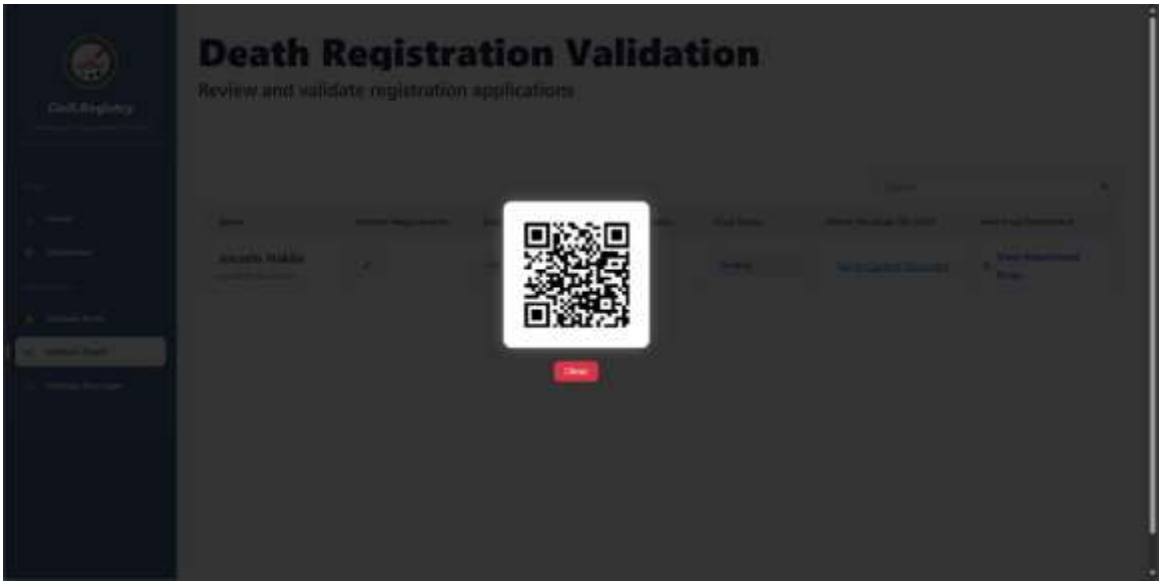


Figure 31.1: Death Registration QR code

Figure 31.1 illustrates the QR code display interface of the Death Registration Validation module in the Civil Registry Management System. The system generates a unique QR code for each birth registration record. When this QR code is scanned using the camera, it automatically redirects the user to the CS Scanner page or module, allowing staff to verify and capture the corresponding registration document. A “Close” button is provided below the QR code to return to the main validation interface.

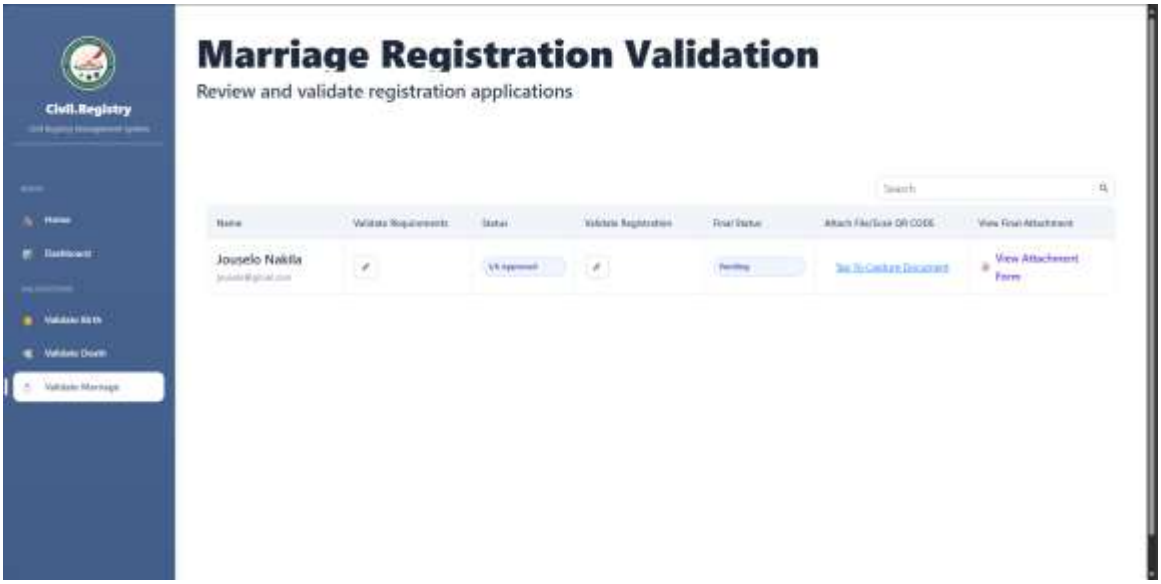


Figure 32: Marriage Registration Validation Interface

Figure 32 shows the Marriage Registration Validation part of the Civil Registry Management System. It lets people who can review and approve marriage registration applications. The interface shows the applicant's name and lets them check their requirements and registration status. The system shows the current status (Pending) and lets you look at the registration form again.



Figure 32.1: Marriage Registration QR code

Figure 32.1 illustrates the QR code display interface of the Marriage Registration Validation module in the Civil Registry Management System. The system generates a unique QR code for each birth registration record. When this QR code is scanned using the camera, it automatically redirects the user to the CS Scanner page or module, allowing staff to verify and capture the corresponding registration document. A “Close” button is provided below the QR code to return to the main validation interface.

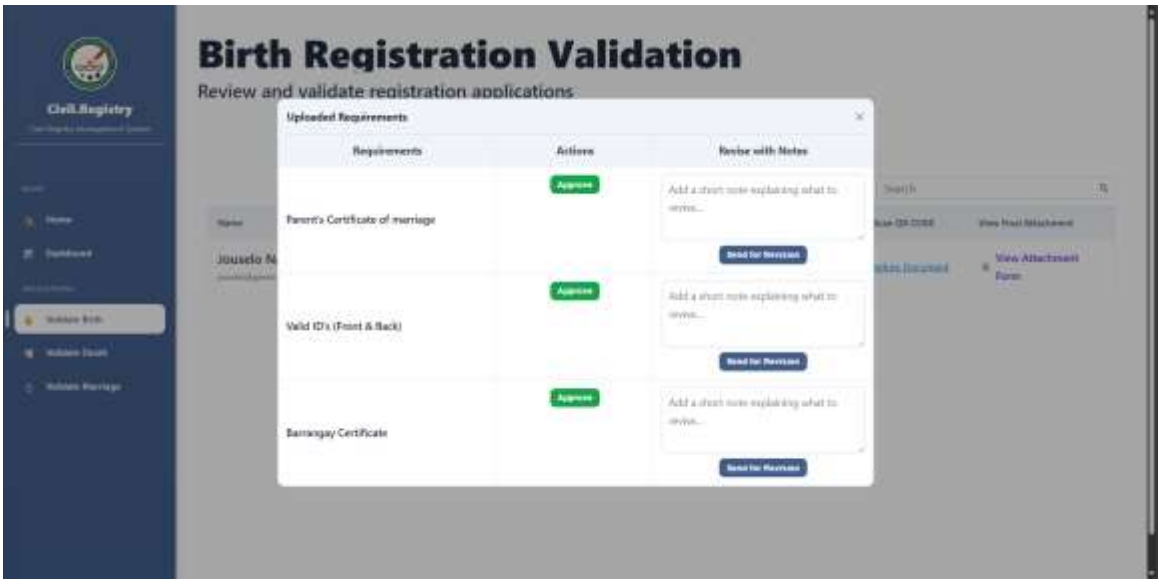


Figure 33: Uploaded Requirements Interface

Figure 33 shows the uploaded requirements submitted by applicants, such as birth certificates and parent identification cards. Staff users can approve the document or request revisions by adding explanatory notes. Once a submission is either approved or marked for requirements revision, the applicant is automatically notified through the system and receives an email notification. The email clearly indicates the status of the submission, whether it has been approved or requires further revision and provides any remarks or instructions from the staff. This functionality ensures accuracy, completeness, transparency, and timely communication in the document validation process.



Figure 34: Certificate of Live Birth Form Interface

Figure 34 shows the interface for the certificate of live birth, which allows users to input personal and birth-related details directly into the digital version of the municipal form. It supports online data entry, verification, and submission, significantly reducing manual paperwork while maintaining the standardized layout required by the Philippine Statistics Authority (PSA).

3.8 Software Platforms, Development Environment and Tools

Table 1

Software Platforms, Development Environments and Tools

Components	Specification	Usage
Internet Browser	Any	A web browser enables users to easily access and display content from the internet. It is responsible for loading the ISO IQA Management System's interface allowing users access to the system's features after a successful login.
Front-end	HTML	HTML defines the structural layout of the ISO IQA Management System's interface arranging elements like forms, texts, and other important elements.
	CSS	CSS is used for improving the aesthetics and designs of the HTML structure providing a more engaging and responsive website.
	JavaScript	To make the website interactive and provide more functionality for the users, JavaScript is used to handle all client-side functionalities.
	Bootstrap	Bootstrap allows developers to easily design responsive and mobile-first web pages by providing a collection of pre-designed components, grids, and styles to provide a more responsive website regardless of the screen size and device.
	JQuery	jQuery is a quick and efficient JavaScript library that simplifies HTML document manipulation, event handling,



		animations, and Ajax interactions. It enables developers to write less code for common web tasks and ensures cross-browser compatibility.
Back-end	MySQL (Database)	MySQL powers data management by providing a robust database, allowing developers to store and manage information efficiently. By utilizing the interface of phpMyAdmin, developers can manage their databases and tables and can also perform queries.
	PHP	PHP is used on the server side which manages the interactions between user inputs and the database ensuring smooth data flow.
	PHPMailer	PHPMailer is a popular open-source PHP library used to send emails securely and reliably from PHP applications. It provides a simple interface for sending emails via SMTP, PHP's mail() function, or other mail transport methods.
Server	Apache	Utilizing the XAMPP control panel, developers can manage the behavioral state of its components ensuring smooth and easy access to local host servers like Apache and MySQL.

3.9 Hardware Requirements

Table 2

Hardware Requirements

Components	Specification	Usage
Device	RAM (16 GB)	RAM facilitates fast and temporary data storage. It is used to enhance the speed and performance of devices in accessing information.
	Processor (Intel Core i5)	The processor or the CPU, acts as the central component of the computer, handling the execution of instructions sent from the website. It also facilitates the processing of data.
	SSD (128 GB)	Devices employ SSD for storing large amounts of audit files, system software, and reports, while flash storage is commonly used in mobile devices for lightweight data storage. These storage solutions ensure quick retrieval and secure long-term data handling.
	Flash Storage (8 GB)	
Printer	Any Ink Jet Printer Units	Inkjet printers are utilized for printing audit reports, data summaries, and compliance documents, producing clear and high-quality output. These printers are versatile and affordable, making them suitable for regular printing tasks related to audit management, such as generating official documents for internal use or distribution to stakeholders.

3.10. Research Ethical Standard

The researchers will ensure that the study, "Online Transaction System of Civil Registrar Office of Buenavista Agusan Del Norte," is conducted in full adherence to ethical standards, data privacy, and the rights of all participants. During the research process, we will adhere to the following ethical guidelines.

A. Protection of Intellectual Property Rights (IPR)

The researchers ensured that everything in this project complies with the rules on intellectual property rights. The Online Transaction System for the Civil Registrar's Office of Buenavista was developed independently by the researchers. The Philippine Copyright Law, RA 8293, protects the original source code, database, and system design. The researchers also acknowledge that certain system features may be eligible for patent protection and that the completed system may fall under institutional ownership, depending on the policies of the Civil Registrar's Office and the academic institution. Proper credit was given to all external resources used; the researchers utilized open-source software and reference materials in compliance with their respective license conditions and cited them accordingly. Trademarks used in this project, such as the official logos of the LCR Office and LGU Buenavista were used only for system representation and remain the property of their respective institutions. The researchers uphold ethical standards by respecting the work of others and recognizing all sources used.

B. Informed Consent

Before the system testing and evaluation, the participants mainly LCR staff and selected citizens were clearly informed about the purpose of the study, the procedures involved, and any potential risks and benefits. The ethical briefing was explained in language that the participants

understood to ensure clarity. They received written consent forms that contained information on study objectives, participation requirements, confidentiality assurances, and their right to withdraw at any time without penalty. All participants read and signed the consent forms before joining the study. No animals were involved, so animal ethics approval was not required.

C. Data Privacy and Confidentiality

The study adhered to the provisions of the Data Privacy Act of 2012 (RA 10173) to maintain the confidentiality and security of all collected information. The researchers ensured that no personally identifiable information was shared, and anonymization was implemented by removing or coding personal details before analysis. Electronic records were secured through restricted access and data-protection measures, including the hashing of sensitive information stored within the system. Printed materials were placed in secure storage accessible only to the research team. For academic purposes, any sharing of anonymized data required participant consent and institutional approval. All data were stored only for the duration necessary for the study and were deleted or disposed of properly after use.

D. Voluntary Participation

Participation in this study was entirely voluntary. The participants were clearly informed about the nature and purpose of the study and were made aware that they could withdraw at any time without any negative consequences. To maintain neutrality and ensure true voluntariness during the testing and feedback process, the research team did not use any form of coercion or provide incentives.

E. Minimization of Harm

The study determined that the system posed no physical, emotional, or psychological risks to the participants. Prior to the system evaluation, the researchers explained the purpose of the activity and demonstrated how to use the interface to minimize confusion or technical difficulties. Participants were informed that they could request assistance at any time or stop their involvement if they experienced discomfort or uncertainty during the testing process. Since the study did not involve animals, the 3Rs principle (Replacement, Reduction, and Refinement) did not apply.

F. Beneficence

The study contributes positively to society by supporting improved public service delivery and promoting e-governance through a more efficient Online Transaction System for the Civil Registrar's Office. Although participants may not receive direct personal benefits, their involvement helps enhance system usability and overall service quality. Research findings and results will be made available to participants upon request, in coordination with the Civil Registrar's Office.

G. Justice and Fair Participant Selection

Participants were selected fairly based on their roles and relevance to the civil registration process, including staff members from the Local Civil Registrar's Office and users who had prior experience with manual registration procedures. No participant was excluded on the basis of gender, socioeconomic background, ethnicity, or status. The selection process upheld the principles of equal opportunity and non-discrimination, ensuring fair representation and the absence of bias throughout the research process.

H. Data Integrity and Accuracy

To ensure data reliability, the researchers employed standardized testing and evaluation methods aligned with the ISO 25010 software quality standards. The research team systematically collected data and validated it through multiple review stages to maintain accuracy and consistency. AI tools such as ChatGPT assisted only with content organization and technical writing support, while Grammarly was used for grammar refinement; however, all AI-assisted outputs were manually reviewed and verified by the researchers to prevent errors or biases. Documented limitations, such as the sample size, the scope of system testing, the restriction of testing to a single municipality, and the absence of an online payment feature, were acknowledged transparently in the study.

I. Transparency in Reporting

All findings, results, and analyses are presented truthfully without any form of data manipulation or fabrication. The researchers upheld academic integrity and ethical reporting throughout the study. All copyrighted, referenced, and acknowledged materials, including external sources, and tools used in system development, are properly cited in the reference section.

J. Use of Patented or Copyrighted Materials

The system exclusively utilized open-source or appropriately licensed resources, including PHP libraries, MySQL database systems, and Bootstrap templates. No patented technologies were used in the development of the system. All external materials were cited in accordance with copyright requirements, ensuring proper attribution, full compliance with licensing conditions, and the absence of any copyright infringement.

K. Ethical Considerations for Animals and Human Trials

The study involved human participants, including LCR staff and community respondents, who were informed about the nature of the study, its minimal risks, and its potential benefits. Participant welfare was protected by ensuring that they could request assistance or withdraw from the activity at any time without consequences, and by avoiding any procedures that could cause discomfort. Their anonymity and privacy were maintained by removing personal identifiers and handling all collected information confidentially. No animals were involved in the project; therefore, ethical clearance for animal use was not required.

L. Responsible Use of AI and Related Technologies

AI tools, such as ChatGPT, were used responsibly and only for concept organization, grammar refinement, and document enhancement. These tools were not used to generate research data, conclusions, or results, thereby preventing any form of plagiarism, fabrication, or bias in the study. All AI-assisted outputs were manually reviewed and validated by the researchers to maintain accuracy, ensure ethical use, and preserve the integrity of the research process.

M. Ethical Approval

The study obtained ethical clearance from the Saint Michael College of Caraga Institutional Research Ethics Committee prior to implementation. The approval, issued on July 26, 2025, confirmed compliance with institutional and national ethical standards concerning data privacy, intellectual property, and participant protection.

System Evaluation

This section describes the different evaluation procedures done to the system to assess its functional suitability, performance efficiency, and usability. The survey instrument is adapted from ISO 25010 Software Product Quality Standards; the test scale gives four options to the respondents and will rate the system based on the question with 4 as the highest and 1 as the lowest.

Data were analyzed and interpreted based on the following parameter

Table 3

Functional Suitability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional
3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Table 3 presents the criteria used to evaluate a product or system's functional suitability according to the ISO survey.

Table 4
Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the system's overall performance efficiency in executing its functions and tasks.

Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable
1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to the usability of the system.

Table 6

Respondent's Distribution

SMCC Learning Resource Center		
Position	**N (Population)**	**Percentage (%)**
Personnel	1	12.5%
Student	7	87.5%
Total	**8**	**100**

Table 6 outlines the distribution of respondents who participated in evaluating the system within the SMCC Learning Resource Center. The respondents are categorized based on their positions, with 12.5% holding personnel roles and 87.5% being students. This indicates that most feedback comes from students directly interacting with the system. Since they are the primary users, their insights provide a more accurate assessment of the system's functionality, efficiency, and usability in real-world scenarios.

CHAPTER 4

SUMMARY, CONCLUSION, RECOMMENDATION,

4.1. Summary of Findings

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Membership/Affiliations

ORGANIZATION	POSITIONS HELD
None	**None**

APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo
B - Instrument with Informed Consent
C - Map of the Research Locale *(Include Coordinates)*
D - Curriculum Vitae
E - Narrative & Photo Documentation
F- Ethics Committee Approval

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